2020 IEEE Intelligent Vehicles Symposium (IV)
October 20-23, 2020. Las Vegas, USA

Connected and Autonomous Vehicles cooperate with the pedestrian in industrial sites based on trajectory optimization and vehicle signalization system.

Meng Zhang, Abdeljalil Abbas-Turki, Alexandre Lombard and Abderrafiaa Koukam

Abstract— **Connected and autonomous vehicles (CAV) is the development trend in the field of transportation systems. Recent studies show that the resources sharing between pedestrians and CAV is a big challenge. Considering traffic safety and efficiency at that sharing point not only requires a collision avoidance system but also more communicative behaviors of the CAV. More precisely, pedestrian needs to understand the intention of the incoming CAV whether it will cross first or not according to its speed profile. This paper uses the optimal trajectory control to provide CAV with a communicative behavior. A scenario where CAV and pedestrian cooperate together to cross a conflict zone is studied. A communicative CAV behavior is designed through an objective function. Hamiltonian analysis is used to derive the optimal control for the CAV. Based on Oculus virtual reality platform, the proposed approach is tested and the behavior of cars and pedestrians are studied. The tests show that this approach provides CAVs with a kind of automatic courtesy.**

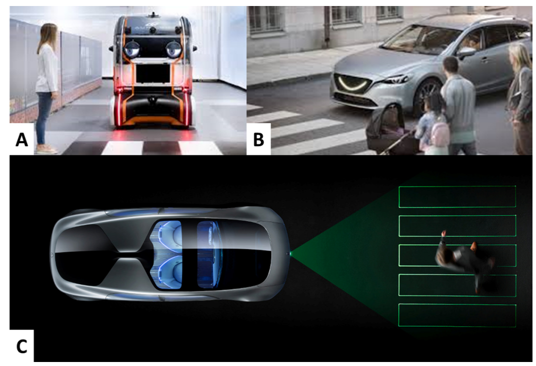

Fig. 1. Signalisation system at the CAV level: A- The CAV informs the pedestrian that she is detected -B A smile pictogram displayed at the CAV's grill -C A pictogram projected on the road

I. INTRODUCTION

Connected and autonomous vehicles (CAV) technology allows to improve safety and efficiency of the transportation system in terms of travel time and riding comfort [1][2]. Transportation in closed industrial sites provides a conducive environment for autonomous and connected vehicle technologies. Indeed, the scope of operation can be extended gradually, by allowing interactions with other road users. One important step to gain autonomy in industrial sites is collision avoidance with obstacles [3]. However this is not enough to behave with humans at intersection points. Indeed, they seek for a communicative behavior in order to dynamically interact with CAV and find a common optimal solution. This needs techniques that optimise interaction between road users and CAV. Many researches have been done recently to answer this issue as presented in [4][5][6].

In the paper, the study area of interest is unsignalized crosswalk. For example A pedestrian wants to cross a road without lights and zebra crossings. A part of the solution to this problem is a signalization system at the CAV level as showed in Fig. 1. The CAV emits a signal that can be easily interpreted by pedestrians, such as a pictogram displayed on the CAV's grill and/or projected on the road near the pedestrian. This informs the pedestrian first that she/he is detected by the CAV and second, about the intention of the vehicle. This allows an explicit communication with the pedestrian. The other part of the solution lies in the decision process that determines who goes first and accordingly computes the suitable speed profile of the CAV.

The speed profile is not neutral [7], [8]. It should provide a communicative behavior for confirming the intention of the CAV. The speed profile is the implicit communication that should not contradict the signalization, i.e, explicit communication. For instance, a CAV that displays the green but accelerates to resort at the last seconds to the emergency brake is not a suitable behavior.

To design the speed profile, this paper considers a CAV which interacts with the pedestrian in order to optimise with her/him a given objective function. The choice of the function must fit the pedestrian behavior. Unfortunately, studying the suitable function faces many challenges. The most important one is to overcome the non-linearity of the occupancy constraint that prevents from getting an explicit solution.

Such an approach is recently investigated for cooperative CAV at intersections. Because, the studied systems are limited to robots, it is hard to extend the results directly to the human-robot interaction. Many works avoid the non-linearity by studying a distributed problem where the sequence and consensus times are predefined [9]. However, this prevents from addressing the consensus issue. One notable work is presented in [10]. The authors developed a distributed coordinated algorithm to obtain real-time CAV trajectories. The algorithm can only be applied to find a consensus between CAVs since the speed profile is computed by re-

*This work is supported by China Scholarship Council

The four authors are with Connaissance et Intelligence Artificielle Distribuées, Université de Bourgogne Franche-Comté UTBM, UTBM (Site de Belfort), 90010 Belfort Cedex, France meng.zhang@utbm.fr

978-1-7281-6673-5/20/$31.00 ©2020 IEEE

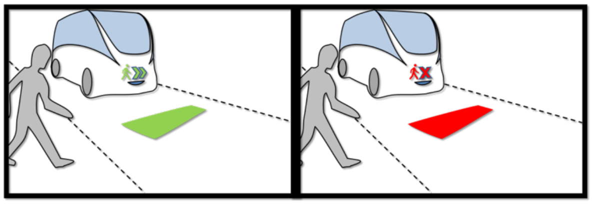

Fig. 2. CAV interacting with pedestrian for sharing the conflict space.

laxing the conflict constraint. To overcome the non-linearity, [11] formulates the problem in space coordinates, rather than time. This considerably simplifies the conflict constraint formulation. However, because of using space coordinates, the state variable $z(p) = 1/v(p)$ (with p and v being the position and the speed, respectively), prevents from studying cases when the speed equals zero. To provide a solution of the modeling problem, we use sigmoid approximation of the discrete occupation constraint.

This article uses a new experimental method-based on Oculus Quest which is a virtual reality platform. The reason is that pedestrian behavior is affected by subjective will and also has complex characteristics. She/he is difficult to simulate. Using Oculus Quest allows real people to enter virtual scenes. In this way, it not only guarantees the authenticity of the experimental data, but also has the characteristics of security.

This paper is organised as follows. First, it introduces the studied conflict problem in section II. The section III applies the Hamiltonian analysis to the problem. A numerical example (section IV) is presented and later discussed according to test data (section V). The data of two scenarios (with and without optimal speed profiles) are compared and analyzed.

II. Problem Statement

Let us consider a CAV that is coming at its desired speed ($\widetilde{v}_v$), and also a pedestrian who wants to cross the road. As the vehicle, the pedestrian also has a desired speed ($\widetilde{v}_p$). Therefore, according to their desired speed, the Conflict Zone (CZ) needs to be shared by the pedestrian and the CAV, as shown in Fig. II. The shared space (CZ) is the colored rectangle. Both must adjust their speed to avoid collision. The dynamic of both is formulated as follows:

$$\dot{X}(t) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} X(t) + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} U(t) \tag{1}$$

Such that in (1), $X^T(t) = [d_v(t), v_v(t), d_p(t), v_p(t)]$ and $U^T(t) = [u_v(t), u_p(t)]$, with $d_k(t)$, $v_k(t)$ and $u_k(t)$ being the traveled distance, the speed and the acceleration of the agent $k \in \{v, p\}$, respectively. When $k = v$, it denotes the vehicle whereas if $k = p$, it denotes the pedestrian.

Speed and acceleration constraints are considered as follows:

$$u_k(t) - \overline{u_k} \leq 0 \tag{2}$$
$$\underline{u_k} - u_k(t) \leq 0 \tag{3}$$
$$v_k(t) - \overline{v_k} \leq 0 \tag{4}$$
$$0 - v_k(t) \leq 0 \tag{5}$$

$\overline{u_k}$ and $\underline{u_k}$ designate the maximum and the minimum acceleration (deceleration), respectively. The maximum speed is $\overline{v_k}$. Equation (5) avoids both pedestrian and CAV to move back but allows them to stop. $\overline{u_k}$, $\underline{u_k}$ and $\overline{v_k}$ are defined not only according to physical limitation but also by taking into account criteria such as the safety and convenience of passengers and good on the autonomous shuttle and maintenance costs.

Besides, both CAV and pedestrian must be able to have an exclusive access to the conflict space. To this end, we consider a counter function $f_{k,i}(t)$, with with $i \in \{enter, exit\}$ that gives the number of times the event i happened until date t. Since, k designates only one agent that is either a CAV or a pedestrian, $f_{k,i}(t)$ equals either 0 or 1 for all (k,i). In order to associate, the counter functions $f_{k,i}$ with the dynamic of both pedestrian and CAV, the minimum traveled distances to validate the event i from the origin (0) is defined. $d_{enter,k}$ is the distance that agent needs to travel to get into CZ whereas $d_{exit,k}$ includes the length of k in such a way the CZ is totally cleared. Hence, we have:

$$f_{k,i}(t) = \begin{cases} 0 & \text{if } d_k(t) < d_{k,i} \\ \\ 1 & \text{if } d_k(t) \geq d_{k,i} \end{cases} \tag{6}$$

From [12], the access to CZ is safe if the following constraint holds:

$$f_{p,enter}(t) - f_{p,exit}(t) + f_{v,enter}(t) - f_{v,exit}(t) - 1 \leq 0 \tag{7}$$

According to (6), $f_{k,i}$ is a discrete function that is not differentiable at $d_k(t) = d_{k,i}$. For this purpose, we consider the following sigmoid approximation:

$$f_{k,i}(t) \simeq fs_{k,i}(t) = \frac{1}{1 + e^{\lambda(d_{k,i} - d_k(t))}} \tag{8}$$

$fs_{k,i}$ have the following derivative function:

$$\frac{\partial fs_{k,i}}{\partial d_k} = \lambda fs_{k,i}(t)\left(1 - fs_{k,i}(t)\right) \tag{9}$$

In (8), as λ get bigger, the sigmoid function $fs_{k,i}$ get closer to the discrete behavior of $f_{k,i}$. In order to keep CZ without overlapping movement when using $fs_{k,i}$, a safety distance sd is included according to the value of λ. In the following we assume that λ has a very large value. It results from (9):

$$\lim_{\lambda \to \infty} \frac{\partial fs_{i,k}}{\partial d_k} \to \begin{cases} 0 & \text{if } d_k(t) \notin [d_{k,i} - ds, d_{k,i} + ds] \\ & \quad \text{with } ds \to 0 \\ +\infty & \text{if } d_k(t) \to d_{k,i} \end{cases} \tag{10}$$

Indeed, if $d_k = d_{k,i}$ then $\frac{\partial fs_{i,k}}{\partial d_k} = \frac{1}{4}\lambda$. In the following, $fs_{i,k}$ with a very large λ is used instead of $f_{i,k}$. To be able to extract a communicative behaviour, it remains to define the objective function. The function must match as well as possible the behavior and the understanding of the pedestrian when she/he crosses the road. Many functions can be tested. Without loss of generality, this paper considers the minimization of the following objective function:

$$J(u_v,u_p) = \int_0^{t_f} (v_v(t)-\widetilde{v}_v)^2\,dt + w_p\int_0^{t_f}(v_p(t)-\widetilde{v}_p)^2\,dt \quad (11)$$

$w_p \geq 1$ is the weighting factor of the pedestrian and t_f is the time horizon of optimization. $w_p \in [1,+\infty]$ acts as cursor to give more or less advantage to pedestrian according to the policy of the industrial site. t_f is defined to allow both agents to regain their desired speed after they cross CZ. $t_f = \max(t_{v,f}, t_{p,f})$ with $t_{k,f}$ is the time needed for an agent k to recover $\widetilde{v}_p$. One can note from the definition of the objective function in (11), that $J(u_v,u_p)$ penalizes the deviation from the desired speed of each user of CZ during t_f. t_f is computed according to the speed profile presented in section IV. In the following, we solve the problem in general case without giving values to neither to t_f nor to w_p. A discussion about the value w_p will be provided in section VI, whereas $t_{k,f}$ can be easily computed from the optimal trajectory.

III. Optimal trajectory

In order to obtain the optimal trajectory analytically for real time applications, we use a Hamiltonian analysis of the problem. In this analysis, we consider that when the CAV detects a pedestrian as it enters the control zone. We consider here that both CAV and pedestrian cooperates to minimize $J(u_v,u_p)$. According to the dynamic equations (1) and the constraints of accelerations (2 and 3) and of speeds (4 and 5), the Hamiltonian function is formulated as follows:

$$\begin{aligned} H\big(t,x(t),u(t)\big) &= \big(v_v(t)-\widetilde{v}_v\big)^2 + w_p\big(v_p(t)-\widetilde{v}_p\big)^2 \\ &+ \sum_{k\in\{v,p\}} \Big(\lambda_k^d v_k(t) + \lambda_k^v u_k(t) \\ &+ \mu_{1,k}\big(u_k(t)-\overline{u_k}\big) + \mu_{2,k}\big(\underline{u_k}-u_k(t)\big) \\ &+ \mu_{3,k}\big(v_k(t)-\overline{v_k}\big) + \mu_{4,k}\big(0-v_k(t)\big)\Big) \\ &+\mu_5\Big(\sum_{k\in\{v,p\}}\big(fs_{k,enter}(t)-fs_{k,exit}(t)\big)-1\Big) \end{aligned} \quad (12)$$

Where λ_k^d and λ_k^v are the costates and $\mu_{1,k}$, $\mu_{2,k}$, $\mu_{3,k}$, $\mu_{4,k}$ and μ_5 are the Lagrange multipliers. with :

$$\mu_{1,k} = \begin{cases} >0 & \text{if } u_k(t) = \overline{u_k} \\ 0 & \text{if } u_k(t) \neq \overline{u_k} \end{cases} \quad (13)$$

$$\mu_{2,k} = \begin{cases} >0 & \text{if } u_k(t) = \underline{u_k} \\ 0 & \text{if } u_k(t) \neq \underline{u_k} \end{cases} \quad (14)$$

$$\mu_{3,k} = \begin{cases} >0 & \text{if } v_k(t) = \overline{v_k} \\ 0 & \text{if } v_k(t) \neq \overline{v_k} \end{cases} \quad (15)$$

$$\mu_{4,k} = \begin{cases} >0 & \text{if } v_k(t) = 0 \\ 0 & \text{if } v_k(t) \neq 0 \end{cases} \quad (16)$$

$$\mu_5 = \begin{cases} >0 & \text{if an agent } k \text{ is in CZ} \\ 0 & \text{if CZ is empty} \end{cases} \quad (17)$$

The Euler-Lagrange equation $\frac{\partial H}{\partial x} = \dot{\lambda}$ becomes:

$$\begin{aligned} \dot{\lambda}_k^d &= \frac{\partial H}{\partial d_k} \\ &= \mu_5(t)\lambda\Big(fs_{k,enter}(t)\big(1-fs_{k,enter}(t)\big) \\ &\quad -\lambda fs_{k,exit}(t)\big(1-f_{k,exit}(t)\big)\Big) \end{aligned} \quad (18)$$

and:

$$\begin{aligned} \dot{\lambda}_v^k &= \frac{\partial H}{\partial v_k} \\ &= 2\cdot w_k v_k(t) - 2\cdot w_k\widetilde{v}_k + \lambda_d^k(t) + \mu_{3,k}(t) - \mu_{4,k}(t) \end{aligned} \quad (19)$$

with $w_v = 1$.

From (18) and (10), λ_k^d is formulated as follows:

$$\lambda_d^k(t) = \begin{cases} a^{b,k} & \text{if agent } k \text{ is } \textbf{b}\text{efore CZ,} \\ a^{i,k} & \text{if agent } k \text{ is } \textbf{i}\text{n CZ,} \\ a^{a,k} & \text{if agent } k \text{ is } \textbf{a}\text{fter CZ,} \end{cases} \quad (20)$$

with $a^{b,k}$, $a^{i,k}$ and $a^{a,k}$ being constants. In the following, we put $\lambda_p^k(t) = a^{j,k}$, with $j \in \{b,i,a\}$ and the value of j depends on if the agent k is before or after CZ.

The necessary condition of the optimality is $\frac{\partial H}{\partial u_k} = 0$, which gives the following result:

$$\begin{aligned} \frac{\partial H}{\partial u_k} &= \lambda_v^k(t) + \mu_{1,k}(t) - \mu_{2,k}(t) = 0 \\ \lambda_v^k(t) &= \mu_{2,k}(t) - \mu_{1,k}(t) \end{aligned} \quad (21)$$

From (21), (13) and (14), $\lambda_v^k(t)$ is formulated as follows:

$$\lambda_v^k(t) = \begin{cases} -\mu_{1,k}(t) & \text{if } u_k(t) = \overline{u_k} \\ \mu_{2,k}(t) & \text{if } u_k(t) = \underline{u_k} \\ 0 & \text{otherwise} \end{cases} \quad (22)$$

From the above analysis, the obtained formulation of the speed profile of the agent k is as follows:

$$v_k(t) = \widetilde{v}_k + \frac{-a^{j,k} - \mu_4(t) + \mu_5(t) - \dot{\mu}_{1,k}(t) + \dot{\mu}_{2,k}(t)}{2\cdot w_k} \quad (23)$$

The analysis of (23) leads to the speed profiles depicted in Fig. 3. Indeed, (23) gives us an expression of $v_k(t)$ that varies according to the constraints of speed and acceleration as well as to the position of the agent. Simply speaking, (23) expresses that the agent k must either keeps its speed constant or accelerate and decelerate at either $\overline{u_k}$ or $\underline{u_k}$. Because of $a^{j,k}$, this speed variations depend on if the agent is before, inside or after the conflict zone. The optimal speed profiles presented in Fig. 3 depends on whether or not the agent k crosses the conflict zone first. More precisely, both agents find a consensus time t_c that minimizes $J(u_v,u_p)$. The leader accelerates ($\overline{u_k}$) during $t_{a,l}$ and keeps a constant high speed

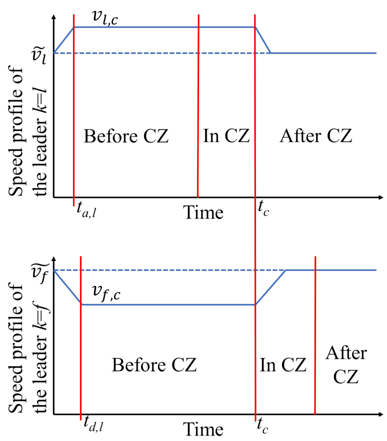

Fig. 3. Optimal speed profiles of the leader and the follower

$v_{l,c}$ to clear CZ at the consensus time, whereas the follower decelerates ($\overline{u_k}$) during $t_{d,f}$ and keeps a low constant speed $v_{f,c}$ to reach CZ at the clearance time.

We notice that the curves make both agents act with a kind of "courtesy". The first accelerates to let the second cross sooner. Both behaviours are communicative. The leader agent accelerates and shows its intention to cross CZ first to the other. The same is true for the follower agent who decelerates, meaning that it yields the right of way to the other agent.

We draw the reader attention to the fact that the human behavior is dynamic and adaptive. Hence, a particular attention to the safety conditions requires to be addressed. Indeed, if the CAV is the follower, it must be able to stop if the pedestrian would change suddenly her/his direction in CZ. Moreover, even if the CAV is designed to respect the speed profile, experience has shown that the CAVs are not as controllable as expected ([13] and [14]). To this end, the following buffer distance bd is considered to determine a safe speed profile for the follower:

$$bd = -\frac{v_{f,c}^2}{2 \cdot \underline{u_v}} \tag{24}$$

bd allows the agent to safely stop before CZ. According to bd, the speed profile is easily adapted. Indeed, from Fig. 3, $v_{f,c} = \underline{u_v} \cdot t_{d,f} + \widetilde{v}_v$. Hence, $t_{d,f}$ is obtained by solving the following equation:

$$\begin{aligned}\frac{1}{2}\underline{u_f} \cdot t_{d,f}^2 + \widetilde{v}_f \cdot t_{d,f} + \left(\underline{u_v} \cdot t_{d,f} + \widetilde{v}_f\right)\left(t_c - t_{d,f}\right)\\ = d_{f,enter} + \frac{\left(\underline{u_f} \cdot t_{d,f} + \widetilde{v}_f\right)^2}{2 \cdot \underline{u_f}}\end{aligned} \tag{25}$$

In (26), the left side of the equality gives the traveled distance by the CAV until t_c whereas the right side constraints the distance to allow the agent to stop before CZ. More precisely the right side equals the remained distance to enter CZ minus the buffer distance given in (24). In this quadratic equation, there is a solution only if the $d_{v,enter}$ is long enough to allow the CAV to stop:

$$t_{d,f} = \frac{\widetilde{v}_f - \underline{u_f} t_c - \sqrt{sc}}{-2 \cdot \underline{u_f}} \tag{26}$$

with

$$\underbrace{2 \cdot \underline{u_f}\left(\frac{1}{2}\underline{u_f} t_c^2 + \widetilde{v}_f t_c - 2 \cdot d_{f,enter}\right) - \widetilde{v}_f^2}_{sc} \geq 0 \tag{27}$$

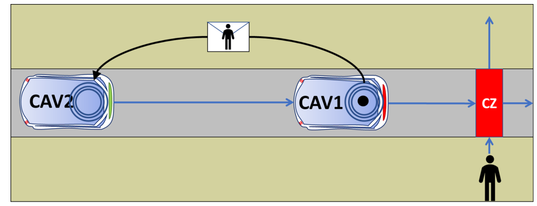

Fig. 4. Example of a platoon of 2 CAV that cooperate with the pedestrian to let her/him cross

IV. Numerical Example

This numerical example bolsters the behaviour of the CAV by extending the conflict presented previously to 3 agents. We consider a platoon of 2 CAVs with a pedestrian who wants to cross the road as presented in Fig. 4. The connectivity of the CAV allows them to communicate together to provide a coherent behavior to the pedestrian. Hence, as shown in Fig. 4, if CAV 1 refuses the request of the pedestrian for safety reasons, it sends a message to CAV 2. The latter will prepare itself, if possible, to allow the pedestrian to cross the road. CAV 1 communicates to CAV 2 the computed t_c that is the time it will clear CZ for letting the pedestrian cross.

TABLE I
Numerical Data

Data	CAV1	CAV2	Pedestrian
$\overline{u_k}$ (m/s^2)	2	2	1
$\underline{u_k}$ (m/s^2)	-3	-3	-1
$\widetilde{v_k}$ (m/s)	10	10	1
$\overline{v_k}$ (m/s)	15	15	1.83 [15]
$d_{k,enter}$ (m)	16	46	1.5
$d_{k,exit}$ (m)	21	51	6
w_k	1	1	20

Numerical data are given in Table IV. In this example, CAV 1, CAV 2 and the pedestrian arrive at their desired $(v_k(0) = \widetilde{v}_k)$ speed. In Fig. 4, CZ is the red rectangle. The initial distance of each agent from the red rectangle is $d_{k,enter}$. $d_{k,exit}$ includes the length of the occupied space by the agent. Finally, the emergency deceleration of CAV equals $-9m/s^2$. The two following pedestrian behaviors are considered:

- **Scenario 1**: The pedestrian behaves as expected by the speed profile
- **Scenario 2**: The pedestrian behaves as expected until she/he is near the end of CZ. She/he stops $0,1m$ before $d_{p,exit}$.

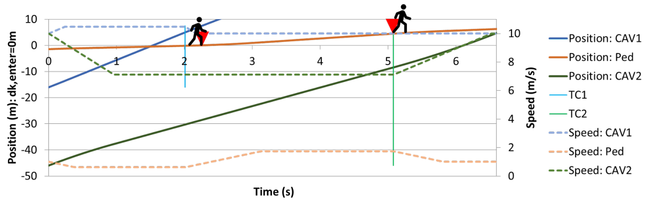

Fig. 5. **Scenario 1**: Optimal trajectories of the three agents : positions and speeds

The optimal solution (scenario 1) is to let CAV 1 cross first. It frees CZ after $2.01s$. The pedestrian is able then to safely access to it. She/he slowdowns first to cross when CZ is cleared. CAV 2 slowdowns too during $1s$, creating a sufficient gap with CAV 1 to allow the pedestrian crossing safely the road. When CZ is cleared by CAV 1, The pedestrian speeds up to free the zone sooner (according to J). CAV 2 moves slowly until the pedestrian leaves CZ. The trajectories of the optimal scenario are presented in Fig. 5. Red triangles indicate when the pedestrian enters to CZ and leaves CZ. One can note that both pedestrian and CAV 2 do not access immediately CZ, in order to be able to stop in the case of unforeseeable behavior of the precedent agent.

Scenario 2 aims to test the safety issue raised by the uncontrollable behavior of the pedestrian. For several reasons, the pedestrian may slowdown and even to completely stop in CZ. She/he also might access to the zone later than expected because of the delayed access of CAV1. Fig. 5 shows the result under extreme conditions in which the pedestrian stops before the end of CZ. Fig. 6 shows that CAV2 immediately reacts to the pedestrian when she/he slowdowns. This figure shows also that the deceleration of CAV 2 doesn't exceed the deceleration bound $\underline{u_{cav2}}$. CAV 2 is thus able to comfortably stop before CZ thanks to (26) and (27).

Finally, Table II compares Scenario 1 (Sc1) with a Spontaneous Behavior (SB) of each agent. In SB, each agent maintains its comfortable state ($\widetilde{v}_k$) whenever it is possible and the pedestrian has the priority if CAV$_i$ can stop at $\underline{u_{cav_i}}$. One can note from this table in line $t_{k,exit}$ that the optimal trajectory allows all agent to cross CZ sooner than in in SB. This is owed to the objective function that:

- distributes the burden of the conflict over all agents: Leaders (CAV 1 and the pedestrian) speed up to let the follower (the pedestrian and CAV 2) access to CZ sooner.
- makes each follower agent maintain the appropriate speed without being necessarily resorted to a complete stop.

V. Experiments and discussion

We performed a first series of experiments using a mixed reality platform (see Fig. 7, Fig. 8, Fig. 9 and Fig. 10). The platform is based on the Oculus Quest, where a real tester physically crosses a virtual road by interacting with a stream of unmanned vehicles: There is no visible driver on the virtual vehicle. The street is a one-way street with two lanes, but now we only consider the easy situation that all cars run only in one lane. Two scenarios are tested. The first one is the spontaneous scenario (see Fig.7 for test and Fig. 8 for the crossing data). The second one is based on the optimal trajectories (see Fig. 9 for tests and Fig. 10 for the crossing data). 42 virtual road crossings have been achieved for each scenario by 28 male and female testers from 7 years old to 62 years old.

In Fig. 7, the vehicles do not emit any signal and their trajectories are not optimised. testers had two different reactions. The first one was to make vehicles dangerously slowdown (35.71%) with many hesitating movements. When the vehicles come to a complete stop, the pedestrian slowly crosses the road (0.), as shown in Fig. 8-A. One can observe that before the dashed red line in this figure the vehicle slows down dangerously ($-9m/s^2$). In the second behaviour (64.28%) the testers wait for a big space between incoming vehicles, quicken the pace (average speed around $1.5m/s$) and even run (more than $2m/s$), in order to avoid a strong deceleration of the vehicle, as shown in Fig. 8-B. Thus, we had two different deceleration ranges: close to $-9m/s^{-2}$ for the first scenario and less than $-3m/s^2$ in the second scenario. Similarly, two range of pedestrian speeds has been observed: near $0.7m/s$ for the former and near $1.6m/s$ for the latter set of behaviors.

Fig. 9 shows the achieved tests of the optimal trajectories. First, the vehicles give a signal (in red or green rectangle) after calculating the optimal speed and assessing the risk (24). This signal is seen by the pedestrians. The first car shares information with the followers if needed (see section IV). The testers observe both signal and behavior of the cars and react accordingly.

In this scenario, all users behave similarly. More precisely, their behavior was close to the recorded crossing data showed in Fig. 10. The pedestrian's average crossing speed is slightly

TABLE II

Comparison between the optimal trajectories and spontaneous ones (SB): $t_{k,exit}$ is the time when the agent k leaves CZ, and $J_k = \int_0^{t_{k,f}} (v_v(t) - \widetilde{v}_k)^2 \, dt$

Agent	CAV1		CAV2		Pedestrian	
Scenarios	Sc1	SB	Sc1	SB	Sc1	SB
$t_{k,exit}$ (s)	2.01	**2.10**	6.63	**9.36**	5.08	**7.03**
$t_{k,f}$ (s)	**2.17**	0	6.52	**12.07**	**5.81**(2.4)	3.11
J_k (m^2/s)	**0.43**	0	40.81	**360.75**	**1.57**(0.28)	0.78

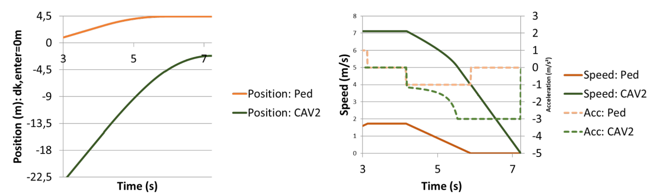

Fig. 6. **Scenario 2**: Trajectory variation according to unforeseeable behavior of the pedestrian : positions, speeds and accelerations

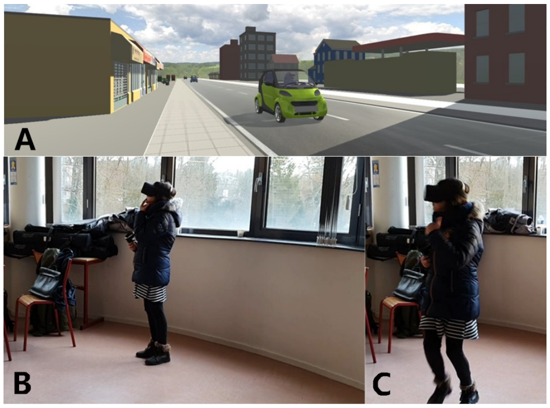

Fig. 7. Mixed reality tests of the pedestrian crossing: A-Observed virtual environment by the testers B-A tester willing to cross the intersection C-The tester quickens the pace while crossing

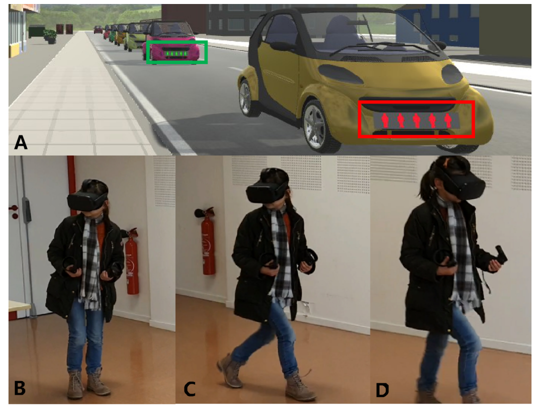

Fig. 9. Mixed reality tests of the pedestrian crossing: A-Observed virtual environment with communicative cars B-A tester observing and waiting C-The tester starting crossing D-The tester crossing "confidently"

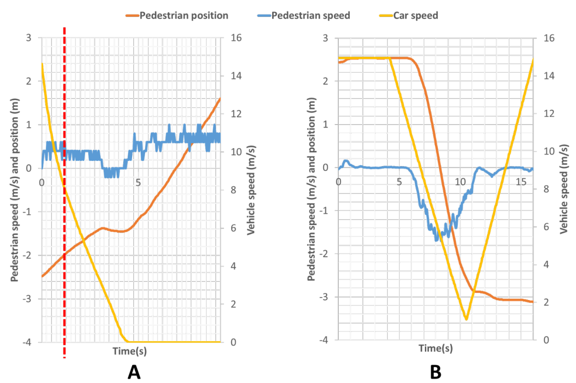

Fig. 8. Crossing data corresponding to scene in Fig. 7

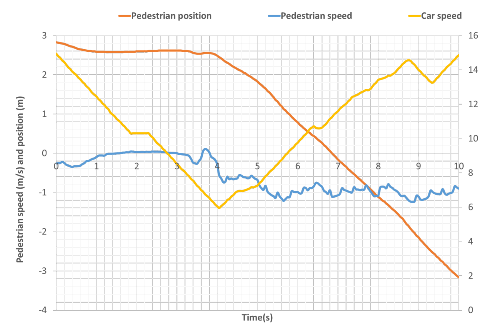

Fig. 10. Crossing data corresponding to scene in Fig. 9

bigger than $1m/s$. In the presented data, the tester observes two different signals that are red for the leader and green for the follower. In the first $4s$, the tester waits because not only the first vehicle gives a red light signal but also she/he needs an amount of time to check both signal and the behavior of the following vehicle (see Fig. 9-B and C). At the same time, the following car (green signal) has started to slow down but because the access time is delayed by the pedestrian to check both behavior and color, the CAV slows down again. In the process of $4s$ $10s$, the vehicle adjusts the driving speed according to the dynamic of the pedestrian in CZ.

We draw the reader attention, that many testers tried to provoke collision. Thanks to (24), no virtual collision has been observed during tests of the optimal scenario and the CAV acceleration has been maintained into the safe range $(ugeq - 3m/s^2)$. This was not the case of the first scenario, where many CAVs resorted to emergency deceleration and some collisions happened. Also, we had a colorblind tester. He successfully crossed the road in the optimal scenario but he needed more time to check the vehicle behavior. This rises the issue of the selected signaling system.

A. Conclusion

Compared with the first experiment, the second experiment uses the proposed interaction behavior, and the curve shows that the pedestrian waiting time before crossing is shorter. This is because the method proposed in this article enables pedestrians to know the "intention" of the car earlier. So the pedestrian decides to pass or wait. In addition, observing the curve of the second experiment, the car continuously adjusts its speed according to the state of the pedestrian, and achieves the purpose of passing "CZ" in the shortest time under the conditions of ensuring safety and acceleration limits.

The proposed approach in this paper seeks to provide a communicative behavior, in addition to vehicle signaling system.The numerical example and experiments based on Oculus Quest show that CAV1 and CAV2 interact with the crossing intention of the pedestrian by creating a comfortable space for crossing. The objective function allows firstly to provide a suitable behavior for both CAVs. Secondly, if the pedestrian reacts as expected, the proposed approach improves the traffic efficiency at industrial sites, avoiding a complete stop. In order to provide a suitable objective and w_p many tests with mixed reality platform are programmed. Also a comparisons with real crossing data in open road are under study.

Since our objective is to propose a CAV cooperative system for pedestrian crossing in industrial sites, we dealt with an optimization problem. The weight chosen for pedestrians can be defined first for safety purpose and second, to find a balance between the duty of CAVs in the industrial site and that of the crew. Many functions can be studied also as the ones for saving CAV energy. Finally, we draw the reader attention that the same approach can be extended to the equilibrium purpose.

The approach deserves to be extended to study more complex environment with several vehicles in both direction and pedestrians (batch of pedestrians). This targets a network of several CAVs that communicate together to find a common solution for a group of pedestrians who intends to cross the road. This raises the challenge of optimization problem as well as of the signaling system.

REFERENCES

[1] D. J. Fagnant and K. Kockelman, "Preparing a nation for autonomous vehicles: opportunities, barriers and policy recommendations," *Transportation Research Part A: Policy and Practice*, vol. 77, pp. 167–181, 2015.

[2] V. Astarita and V. P. Giofré, "From traffic conflict simulation to traffic crash simulation: Introducing traffic safety indicators based on the explicit simulation of potential driver errors," *Simulation Modelling Practice and Theory*, vol. 94, pp. 215–236, 2019.

[3] Y. Luo, P. Cai, A. Bera, D. Hsu, W. S. Lee, and D. Manocha, "Porca: Modeling and planning for autonomous driving among many pedestrians," *IEEE Robotics and Automation Letters*, vol. 3, no. 4, pp. 3418–3425, 2018.

[4] R. Tian, E. Y. Du, K. Yang, P. Jiang, F. Jiang, Y. Chen, R. Sherony, and H. Takahashi, "Pilot study on pedestrian step frequency in naturalistic driving environment," in *2013 IEEE Intelligent Vehicles Symposium (IV)*. IEEE, 2013, pp. 1215–1220.

[5] R. Tian, L. Li, K. Yang, S. Chien, Y. Chen, and R. Sherony, "Estimation of the vehicle-pedestrian encounter/conflict risk on the road based on tasi 110-car naturalistic driving data collection," in *2014 IEEE Intelligent Vehicles Symposium Proceedings*. IEEE, 2014, pp. 623–629.

[6] A. Rasouli and J. K. Tsotsos, "Autonomous vehicles that interact with pedestrians: A survey of theory and practice," *IEEE transactions on intelligent transportation systems*, 2019.

[7] D. Sadigh, N. Landolfi, S. S. Sastry, S. A. Seshia, and A. D. Dragan, "Planning for cars that coordinate with people: leveraging effects on human actions for planning and active information gathering over human internal state," *Autonomous Robots*, vol. 42, no. 7, pp. 1405–1426, 2018.

[8] A. A. Malikopoulos, S. Hong, B. B. Park, J. Lee, and S. Ryu, "Optimal control for speed harmonization of automated vehicles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 20, no. 7, pp. 2405–2417, 2018.

[9] A. A. Malikopoulos, C. G. Cassandras, and Y. J. Zhang, "A decentralized energy-optimal control framework for connected automated vehicles at signal-free intersections," *Automatica*, vol. 93, pp. 244–256, 2018.

[10] A. Mirheli, M. Tajalli, L. Hajibabai, and A. Hajbabaie, "A consensus-based distributed trajectory control in a signal-free intersection," *Transportation research part C: emerging technologies*, vol. 100, pp. 161–176, 2019.

[11] L. Riegger, M. Carlander, N. Lidander, N. Murgovski, and J. Sjöberg, "Centralized mpc for autonomous intersection crossing," in *2016 IEEE 19th international conference on intelligent transportation systems (ITSC)*. IEEE, 2016, pp. 1372–1377.

[12] A. Corréïa, A. Abbas-Turki, R. Bouyekhf, and A. El Moudni, "A dioid model for invariant resource sharing problems," *IEEE Transactions on Systems, Man, and Cybernetics-Part A: Systems and Humans*, vol. 39, no. 4, pp. 770–781, 2009.

[13] M. Quinlan, T.-C. Au, J. Zhu, N. Stiurca, and P. Stone, "Bringing simulation to life: A mixed reality autonomous intersection," in *2010 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2010, pp. 6083–6088.

[14] M. Khayatian, Y. Lou, M. Mehrabian, and A. Shirvastava, "Crossroads+ a time-aware approach for intersection management of connected autonomous vehicles," *ACM Transactions on Cyber-Physical Systems*, vol. 4, no. 2, pp. 1–28, 2019.

[15] H. Hamdane, T. Serre, R. Anderson, and J. Yerpez, "Accident simulation and reconstruction for enhancing pedestrian safety: issues and challenges," 2014.